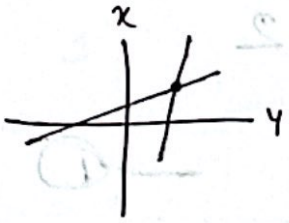
(1)

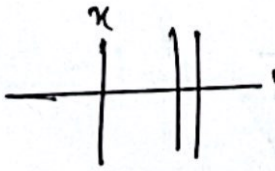
(a)

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

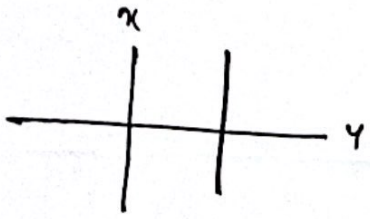
here, Two equations
Two unknowns

- 

if two line intersect at a single point, then there is a unique solution

- 

if two lines are parallel, means if they do not intesect, then there is no solution.



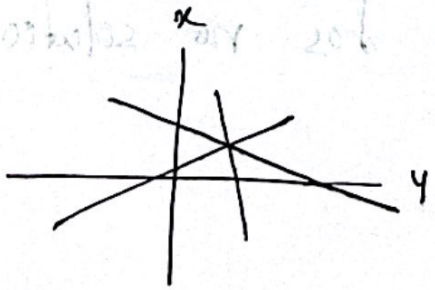


if two lines overlap each other then there are infinately many solution.

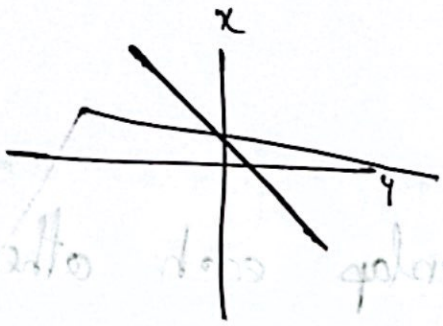
(b)

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

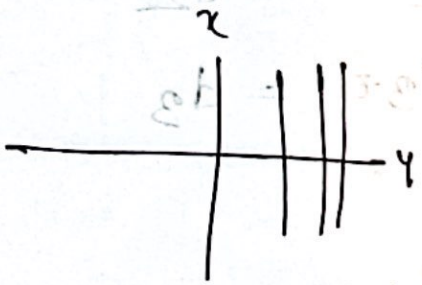
here, three equations
three unknowns



if three planes intercept at a single point, thus there is a unique solution.



if ~~two~~ three planes intercept along a line or overlap, then the system has infinitely many solutions.



if three planes do not intercept, or parallel then it has no solution.

(2)

(a) $x_1 + 2x_2 = -2$

$2x_1 + x_2 + x_3 = 11$

$2x_1 + x_3 = 12$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 2 & 1 & 1 & 11 \\ 2 & 0 & 1 & 12 \end{array}\right) \Rightarrow$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 0 & -3 & 1 & 15 \\ 0 & -4 & 1 & 16 \end{array}\right) \begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 2R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 0 & 1 & -\frac{1}{3} & -5 \\ 0 & 0 & -\frac{1}{3} & -4 \end{array}\right) \begin{array}{l} R_2 = R_2/-3 \\ R_3 = R_3 + 4R_2 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 0 & 1 & -\frac{1}{3} & -5 \\ 0 & 0 & 1 & 12 \end{array}\right) \quad R_3 = R_3 / -\frac{1}{3}$$

Back substitution,

$$x_3 = 12 \qquad \text{——} (1)$$

$$x_2 - \frac{1}{3} x_3 = -5$$

$$\therefore x_2 = \left(\frac{1}{3} \times 12\right) - 5$$

$$= -1 \qquad \text{——} (4)$$

$$\therefore \quad x_1 + 2x_2 = -2$$

$$x_1 = -2 - (2 \times -1)$$

$$= 0.$$

$\therefore \quad x_1 = 0, \; x_2 = -1, \; x_3 = 12 \quad$ (Ans)

$\therefore$ system has unique solution.

(2)

(b)

$$2x_1 + x_2 + 12x_3 = 11$$

$$x_1 + 2x_2 + 9x_3 = -11$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 12 & 11 \\ 1 & 2 & 9 & -11 \end{array}\right)$$

$$-1 \quad 1 \quad 12 \quad 11$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 9 & -11 \\ 2 & 1 & 12 & 11 \end{array}\right) \quad \begin{array}{l} R_1 - R_2 \\ R_2 = R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 9 & -11 \\ 0 & -3 & -6 & 33 \end{array}\right) \quad R_2 = R_2 - 2R_1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 9 & -11 \\ 0 & 1 & 2 & -11 \end{array}\right) \quad R_2 = R_2 \div -3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 5 & 11 \\ 0 & 1 & 2 & -11 \end{array}\right) \quad R_1 = R_1 - 2R_2$$

x_3 is a free variable.

let, $x_3 = t$

$\therefore \quad x_1 + 5t = 11$

$x_1 = 11 - 5t$

$\therefore \quad x_2 + 2t = -11$

$x_2 = -11 - 2t$.

$\therefore x_1 = 11 - 5t, \quad x_2 = -11 - 2t, \quad x_3 = t$

$\therefore$ system has infinitely many solutions depending on t

(3)

(a)

$$\begin{aligned} -x_1 + x_2 + x_3 + 2x_4 &= 11 \\ 2x + 2x_3 + x_4 &= 0 \\ 2x_1 + 2x_3 + x_4 &= 12 \end{aligned}$$

augmented matrix:

$$\left(\begin{array}{cccc|c} -1 & 1 & 1 & 2 & 11 \\ 0 & 2 & 2 & 1 & 0 \\ 2 & 0 & 2 & 1 & 12 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 2 & 1 & 12 \\ 0 & 2 & 2 & 1 & 0 \\ -1 & 1 & 1 & 2 & 11 \end{array}\right) \quad \begin{array}{l} R_1 \leftrightarrow R_3 \\ \\ R_3 \leftrightarrow R_1 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & \frac{1}{2} & 6 \\ 0 & 2 & 2 & 1 & 0 \\ -1 & 1 & 1 & 2 & 11 \end{array}\right) \quad R_1 = R_1/2$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & \frac{1}{2} & 6 \\ 0 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & \frac{5}{2} & 17 \end{array}\right) \quad R_3 = R_3 + R_1$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & \frac{1}{2} & 6 \\ 0 & 1 & 1 & \frac{1}{2} & 0 \\ 0 & 1 & 2 & \frac{5}{2} & 17 \end{array}\right) \quad R_2 = R_2 \div 2$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & \frac{1}{2} & 6 \\ 0 & 1 & 1 & \frac{1}{2} & 6 \\ 0 & 0 & 1 & 2 & 17 \end{array}\right) \quad R_3 = R_3 - R_2$$

Here x_4 is a free variable.

let $x_4 = t$

By back substitution,

$$x_3 + 2t = 17$$

$$\Rightarrow x_3 = 17 - 2t \quad \text{——} (1)$$

$$x_2 + x_3 + \frac{1}{2}t = 0$$

$$\Rightarrow x_2 = -17 + \frac{3}{2}t \quad \text{——} (11)$$

$$x_1 + x_3 + \frac{1}{2}t = 6$$

$$x_1 = -11 + \frac{5}{2}t \quad \text{——— (III)}$$

$\therefore$ $x_1 = -11 + \frac{5}{2}t$, $x_2 = -17 + \frac{3}{2}t$,

$x_3 = 17 - 2t$, $x_4 = t$

$\therefore$ the system has infinitely many solution.

(3)

(b)

$$5x_1 - 5x_2 - 45x_3 + 10x_4 = 10$$
$$-x_1 + x_2 + 9x_3 - 3x_4 = -11$$

augmented matrix

$$\left[\begin{array}{cccc|c} 5 & -5 & -45 & 10 & 10 \\ -1 & 1 & 9 & -3 & -11 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & -1 & -9 & 2 & 2 \\ -1 & 1 & 9 & -3 & -11 \end{array}\right] \quad R_1 = R_1 \div 5$$

$$\left[\begin{array}{cccc|c} 1 & -1 & -9 & 2 & 2 \\ 0 & 0 & 0 & -1 & -9 \end{array}\right] \quad R_2 = R_2 + R_1$$

Back Substitution:



x_2, x_3, [redacted] is a free variable,

let, $x_2 = t_2$ $x_3 = t_3$, [redacted] $(t_n \in R)$

$\therefore x_1 - t_2 - 9t_3 + 2 - t_4 = 2$

$x_1 = [redacted] 2 + t_2 + 9t_3 - 2t_4$ —— (i)

$-t_4 = -9$

$\therefore x_4 = 9$ —— (ii)

$\therefore x_1 = 2 + t_2 + 9t_3 - 18$

$= -16 + t_2 + 9t_3$

$\therefore x_1 = -16 + t_2 + 9t_3$, $x_2 = t_2$, $x_3 = t_3$

$x_4 = 9$ $\therefore$ the system has many solution.

as x_1, x_2 became zero and x_3, x_4 is a free variable, thus,

this system has no solution.

(c)

$$-9x_1 - 18x_2 + x_3 - 2x_4 = 9$$

augmented matrix:

$$\left(\begin{array}{cccc|c} -9 & -18 & 1 & -2 & 9 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 2 & -\frac{1}{9} & \frac{2}{9} & -1 \end{array}\right) \quad R_1 = R_1 \div 9$$

here x_2, x_3, x_4 is free variable.

let, $x_2 = t_2 \quad x_3 = t_3 \quad x_4 = t_4$

$$\therefore x_1 + 2t_2 - \frac{1}{9}t_3 + \frac{2}{9}t_4 = -1$$

$$\Rightarrow x_1 = -1 - 2t_2 + \frac{1}{9}t_3 - \frac{2}{9}t_4$$

$\therefore x_1 = -1 - 2t_2 + \frac{1}{5}t_3 - \frac{2}{5}t_4,$

$x_2 = t_2$

$x_3 = t_3$

$x_4 = t_4$

where $t_2, t_3, t_4 \in \mathbb{R}$

$\therefore$ the system has infinitely many solution.

(4)

$$x_1 + 2x_2 - x_3 = 13$$

$$3x_1 - x_2 + 2x_3 = 31$$

$$4x_1 + x_2 + (\lambda^2 - 9)x_3 = \lambda + 5$$

augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 13 \\ 3 & -1 & 2 & 31 \\ 4 & 1 & \lambda^2-9 & \lambda+5 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 13 \\ 0 & -7 & 5 & -8 \\ 4 & 1 & \lambda^2-9 & \lambda+5 \end{array}\right) \quad R_2 = R_2 - 3R_1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 13 \\ 0 & -7 & 5 & -8 \\ 0 & -7 & \lambda^2-5 & \lambda^2-47 \end{array}\right) \quad R_3 = R_3 - 4R_1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 13 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & \lambda^2-10 & \lambda-39 \end{array}\right) \quad R_3 = R_3 - R_2$$

⇒ for all values except $\lambda = \pm\sqrt{10}$ we will find unique solution.

⇒ for $\lambda = \pm\sqrt{10}$ the system will have no solution.

⇒ for $\lambda = \pm\sqrt{10}$, $\lambda = 39$ the system will have infinitely solution. But it's impossible, so no infinite solutions.

(5)

(a)

$$A = \begin{bmatrix} -1 & -6 & -6 \\ 0 & -3 & -4 \\ 2 & 4 & -7 \end{bmatrix}$$

$$\det(A) = -1 \times \left((-3 \times -7) - (-4 \times 4)\right) - (-6) \times \left(0 \times -7 - (2 \times -4)\right) + -6 \times \left((0 \times 4) - (-3 \times 2)\right)$$

$$= -25$$

as $\det(A) \neq 0$ $\therefore$ the matrix is invertible

Inverted matrix:

$$[A|I] = \left[\begin{array}{ccc|ccc} -1 & -6 & -6 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & 1 & 0 \\ 2 & 4 & -7 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 6 & 6 & -1 & 0 & 0 \\ 0 & -3 & -4 & 0 & 1 & 0 \\ 2 & 4 & -7 & 0 & 0 & 1 \end{array}\right] \quad R_1 = R_1 \div -1$$

$$= \left[\begin{array}{ccc|ccc} 1 & 6 & 6 & -1 & 0 & 0 \\ 0 & -3 & -4 & 0 & 1 & 0 \\ 0 & -8 & -19 & 2 & 0 & 1 \end{array}\right] \quad R_3 = R_3 - 2R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & 6 & 6 & -1 & 0 & 0 \\ 0 & 1 & \frac{4}{3} & 0 & -\frac{1}{3} & 0 \\ 0 & -8 & -19 & 2 & 0 & 1 \end{array}\right] \quad R_2 = R_2 \div -3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 6 & 6 & -1 & 0 & 0 \\ 0 & 1 & \frac{4}{3} & 0 & -\frac{1}{3} & 0 \\ 0 & 0 & -\frac{25}{3} & 2 & -\frac{8}{3} & 1 \end{array}\right] \quad R_3 = R_3 + 8R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 6 & 6 & -1 & 0 & 0 \\ 0 & 1 & \frac{4}{3} & 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & -\frac{6}{25} & \frac{8}{25} & -\frac{3}{25} \end{array}\right] \quad R_3 = R_3 \div -\frac{25}{3}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 6 & 6 & -1 & 0 & 0 \\ 0 & 1 & 0 & \frac{8}{25} & -\frac{19}{25} & \frac{4}{25} \\ 0 & 0 & 1 & -\frac{6}{25} & \frac{8}{25} & -\frac{3}{25} \end{array}\right] \quad R_2 = R_2 - \frac{4}{3}R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 6 & 0 & \frac{11}{25} & -\frac{48}{25} & \frac{18}{25} \\ 0 & 1 & 0 & \frac{8}{25} & -\frac{19}{25} & \frac{4}{25} \\ 0 & 0 & 1 & -\frac{6}{25} & \frac{8}{25} & -\frac{3}{25} \end{array}\right] \quad R_1 = R_1 - 6R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{37}{25} & \frac{66}{25} & -\frac{6}{25} \\ 0 & 1 & 0 & \frac{8}{25} & -\frac{19}{25} & \frac{4}{25} \\ 0 & 0 & 1 & -\frac{6}{25} & \frac{8}{25} & -\frac{3}{25} \end{array}\right] \quad R_1 = R_1 - 6R_2$$

$$\therefore A^{-1} = \left[\begin{array}{ccc} -\frac{37}{25} & \frac{66}{25} & -\frac{6}{25} \\ \frac{8}{25} & -\frac{19}{25} & \frac{4}{25} \\ -\frac{6}{25} & \frac{8}{25} & -\frac{3}{25} \end{array}\right]$$

(b)

$$A = \begin{bmatrix} 3 & -2 & 4 \\ 1 & 0 & -2 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\det(A) = 3 \times (0 - (-1 \times -2)) - (-2) \times ((1 \times 0) - (-2 \times 0)) + 4((1 \times -1) - 0 \times 0)$$

$$= -10$$

$\det(A)$ is not zero. so the matrix A is invertible.

$$[A|I] = \left[\begin{array}{ccc|ccc} 3 & -2 & 4 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & -\frac{2}{3} & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right] \quad R_1 = R_1 \div 3$$

$$= \left[\begin{array}{ccc|ccc} 1 & -\frac{2}{3} & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 0 & \frac{2}{3} & -\frac{10}{3} & -\frac{1}{3} & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right] \quad R_2 = R_2 - R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & -\frac{2}{3} & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & -5 & -\frac{1}{2} & \frac{3}{2} & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right] \quad R_2 = R_2 \div \frac{2}{3}$$

$$= \left[\begin{array}{ccc|ccc} 1 & -\frac{2}{3} & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & -5 & -\frac{1}{2} & \frac{3}{2} & 0 \\ 0 & 0 & -5 & -\frac{1}{2} & \frac{3}{2} & 1 \end{array}\right] \quad R_3 = R_3 + R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & -\frac{2}{3} & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & -5 & -\frac{1}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 1 & \frac{1}{10} & -\frac{3}{10} & -\frac{1}{5} \end{array}\right] \quad R_3 = R_3 \div -5$$

$$\left[\begin{array}{ccc|ccc} 1 & -\frac{2}{3} & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{10} & -\frac{3}{10} & -\frac{1}{5} \end{array}\right] \quad R_2 = R_2 + 5R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & -\frac{2}{3} & 0 & \frac{1}{5} & \frac{2}{5} & \frac{4}{15} \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{10} & -\frac{3}{10} & -\frac{1}{5} \end{array}\right] \quad R_1 = R_1 - \frac{4}{3}R_3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{2}{5} & -\frac{2}{5} \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{10} & -\frac{3}{10} & -\frac{1}{5} \end{array}\right] \begin{array}{l} R_1 = R_1 + \frac{2}{3} R_2 \end{array}$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} & -\frac{2}{5} \\ 0 & 0 & -1 \\ \frac{1}{10} & -\frac{3}{10} & -\frac{1}{5} \end{bmatrix}$$